Piecewise-Defined & Implicit Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.02

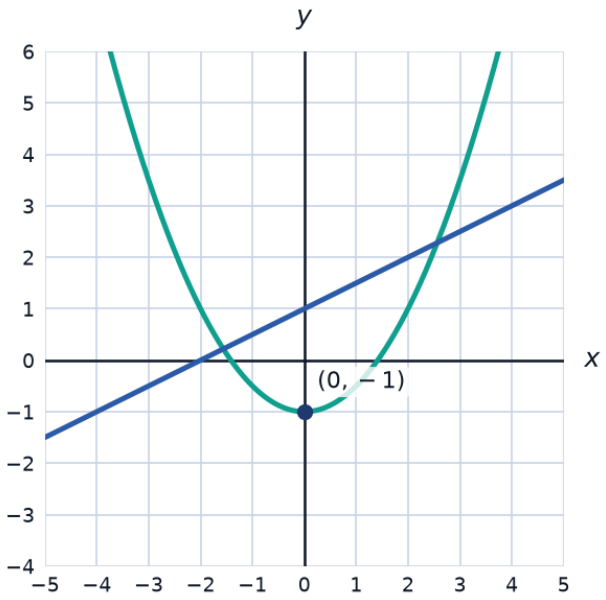
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating piecewise functions

Given $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$, find $f(-3)$ and $f(4)$.

2 [Foundational] • Target: Evaluating piecewise functions

A shipping company charges a flat 5 dollars for packages up to 2 kg, plus 1.50 dollars per kg for anything over 2 kg. Write a piecewise function $C(w)$ for the cost, and find $C(5)$.



3 [Foundational] • Target: Checking continuity of piecewise functions

Find the value of k that makes $f(x) = 3x + k$ for $x < 2$, and $f(x) = x^2 - 1$ for $x \geq 2$, continuous at $x = 2$.

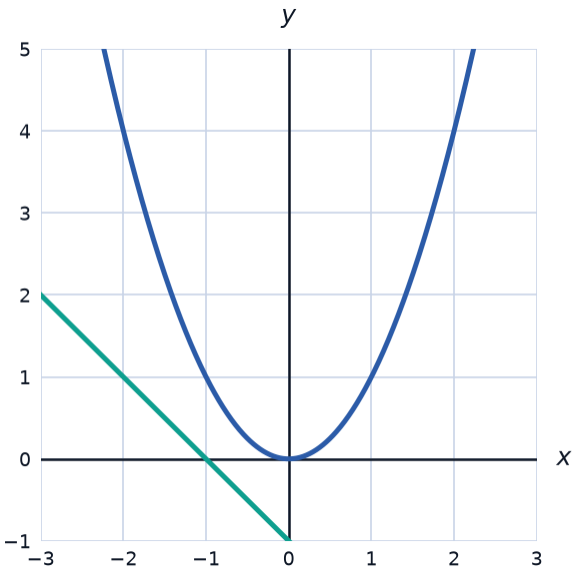
4 [Foundational] • Target: Checking continuity of piecewise functions

Determine whether $g(x) = x + 1$ for $x \leq 1$, and $g(x) = 3x - 2$ for $x > 1$, is continuous at $x = 1$, showing your reasoning.

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5 [Foundational] • Target: Visualizing a piecewise graph

The graph shows $f(x) = -x - 1$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$.

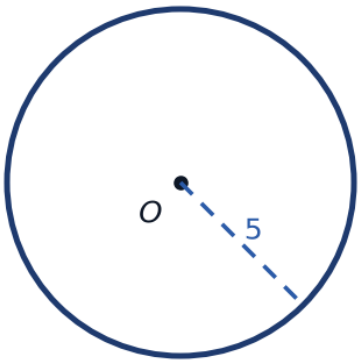


a Evaluate the left piece and the right piece of f at $x = 0$, and use this to determine whether f is continuous there.

b State the range of f shown on the graph.

6 [Foundational] • Target: Implicit relations and testing for functions

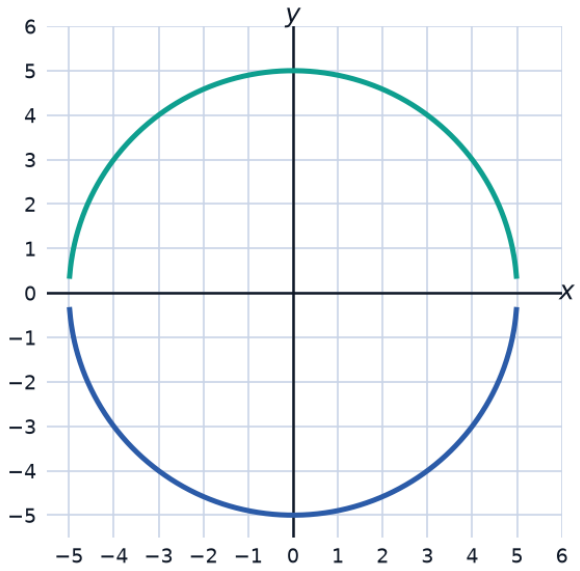
Determine whether $x^2 + y^2 = 25$ defines y as a function of x. Justify using the vertical line test.



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7 [Foundational] • Target: Implicit relations and testing for functions

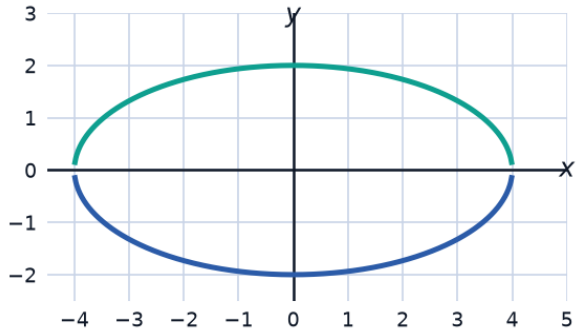
Solve $x^2 + y^2 = 25$ for y explicitly, and state which branch corresponds to a function.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Implicit differentiation-style reasoning (algebraic slope)

For the implicit relation $x^2 + 4y^2 = 16$ (an ellipse), find the two y-values when $x = 2$, and describe what this means about the relation as a function.



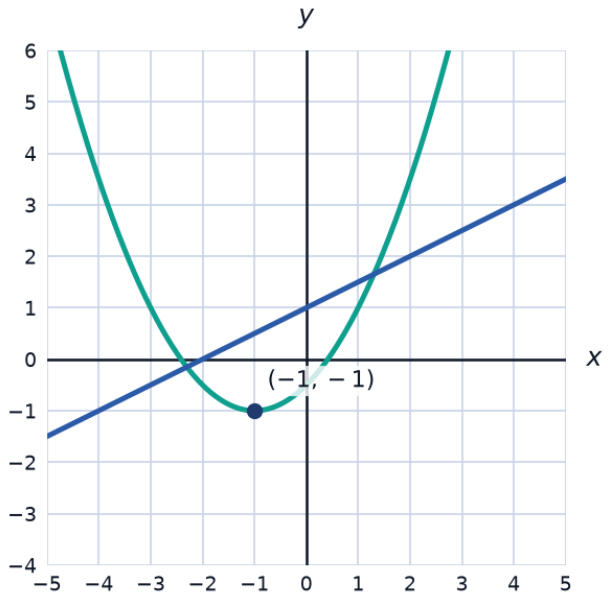
9 [Intermediate] • Target: Modeling with piecewise functions

A tax schedule charges 10% on income up to 10{,}000 dollars and 20% on income above 10{,}000 dollars. Write a piecewise tax function $T(x)$ for income x, and find the tax owed on an income of 25{,}000 dollars.

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10 [Intermediate] • Target: Piecewise functions with three or more pieces

A parking garage charges 3 dollars for the first hour, 2 dollars for each additional hour up to 5 hours total, and a flat 15 dollars for anything over 5 hours. Write this as a three-piece function $P(h)$ for h hours parked, and find $P(4)$.

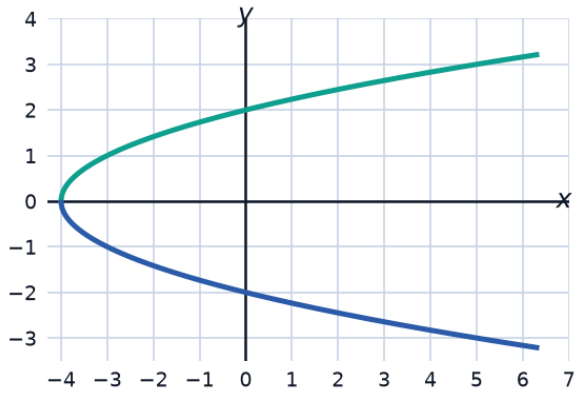


11 [Intermediate] • Target: Piecewise functions with three or more pieces

Evaluate the absolute value function $f(x) = |2x - 6|$ at $x = 1$ and $x = 5$ by first rewriting it as a two-piece function without absolute value bars.

12 [Intermediate] • Target: Graphing implicit relations point-by-point

For the implicit relation $y^2 = x + 4$, find all y-values when $x = 0$, $x = 5$, and $x = -4$, and state the smallest allowed x-value.



13 [Intermediate] • Target: Piecewise functions and average rate of change

For $f(x) = x^2$ on $x < 2$ and $f(x) = 3x - 2$ on $x \geq 2$, find the average rate of change of f on $[0, 4]$.

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14 [Intermediate] • Target: Piecewise functions and average rate of change

Given $g(x) = 2x + 1$ for $x \le 3$ and $g(x) = x^2 - 5$ for $x > 3$, find the average rate of change of g on $[1, 5]$.

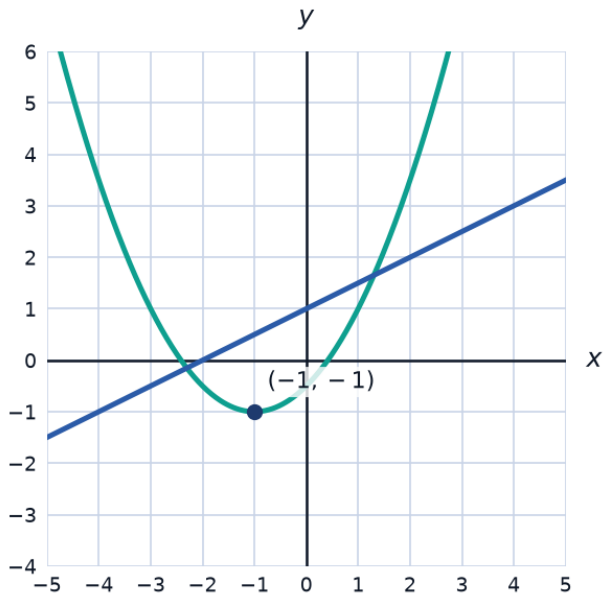
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Piecewise functions and average rate of change • Cross-Lesson: 2.02 Logarithmic Functions & Properties

A cell phone plan charges 20 dollars flat for up to 2 GB of data, plus 8 dollars per GB for each additional GB, up to a maximum monthly charge of 68 dollars (unlimited data cap). Write a three-piece function $M(x)$ for the monthly cost based on x GB used, and find $M(8)$.

16 [Challenging] • Target: Visualizing a piecewise graph • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Find the value of k that makes $f(x) = 7x + k$ for $x < 0$, and $f(x) = 8x - 9$ for $x \ge 0$, continuous at $x = 0$.



17 [Challenging] • Target: Implicit relations and testing for functions • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Find the value of k that makes $f(x) = 9x + k$ for $x < 3$, and $f(x) = -x - 2$ for $x \ge 3$, continuous at $x = 3$.

18 [Challenging] • Target: Implicit differentiation-style reasoning (algebraic slope) • Cross-Lesson: 4.01 Inverse & Composite Functions

Find the value of k that makes $f(x) = -7x + k$ for $x < 3$, and $f(x) = 7x - 8$ for $x \ge 3$, continuous at $x = 3$.

19 [Challenging] • Target: Modeling with piecewise functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

Find the value of k that makes $f(x) = x + k$ for $x < -1$, and $f(x) = 8x + 3$ for $x \ge -1$, continuous at $x = -1$.

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20 [Challenging] • Target: Piecewise functions with three or more pieces • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Find the value of k that makes $f(x) = 8x + k$ for $x < 5$, and $f(x) = -x + 5$ for $x \geq 5$, continuous at $x = 5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating piecewise functions

Given $f(x) = 3x - 5$ for $x < 0$, and $f(x) = -5x + 9$ for $x \geq 0$, find $f(-5)$.

a. 34

b. -19

c. -20

d. -21

22 [Foundational] • Target: Checking continuity of piecewise functions

Given $f(x) = -6x + 4$ for $x < 4$, and $f(x) = 6x + 5$ for $x \geq 4$, find $f(8)$.

a. 53

b. 54

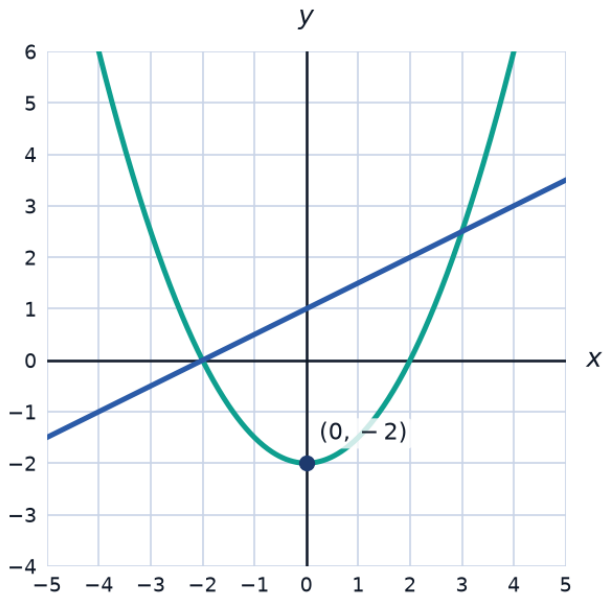
c. -44

d. 52

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23 [Foundational] • Target: Visualizing a piecewise graph

Given $f(x) = -3x + 3$ for $x < 1$, and $f(x) = -6x - 9$ for $x \geq 1$, find $f(1)$.



a. -15

b. -14

c. 0

d. -16

24 [Foundational] • Target: Implicit relations and testing for functions

Given $f(x) = 4x - 7$ for $x < 2$, and $f(x) = 6x - 5$ for $x \geq 2$, find $f(3)$.

a. 14

b. 5

c. 13

d. 12

25 [Foundational] • Target: Implicit differentiation-style reasoning (algebraic slope)

Given $f(x) = 6x + 1$ for $x < -1$, and $f(x) = x - 2$ for $x \geq -1$, find $f(-5)$.

a. -29

b. -30

c. -28

d. -7

26 [Foundational] • Target: Modeling with piecewise functions

Given $f(x) = 2x - 7$ for $x < -1$, and $f(x) = -5x - 4$ for $x \geq -1$, find $f(-6)$.

a. 26

b. -18

c. -19

d. -20

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27 [Foundational] • Target: Piecewise functions with three or more pieces

Given $f(x) = -3x + 8$ for $x < -3$, and $f(x) = -2x + 6$ for $x \geq -3$, find $f(-7)$.

a. 28

b. 30

c. 29

d. 20

28 [Foundational] • Target: Graphing implicit relations point-by-point

Given $f(x) = -2x + 2$ for $x < 4$, and $f(x) = -x - 4$ for $x \geq 4$, find $f(-1)$.

a. 4

b. 5

c. -3

d. 3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Piecewise functions and average rate of change

Find the value of k that makes $f(x) = -2x + k$ for $x < 2$, and $f(x) = 7x - 1$ for $x \geq 2$, continuous at $x = 2$.

a. 17

b. 16

c. 9

d. 18

30 [Intermediate] • Target: Evaluating Piecewise Functions

Find the value of k that makes $f(x) = -9x + k$ for $x < -2$, and $f(x) = -7x + 6$ for $x \geq -2$, continuous at $x = -2$.

a. 38

b. 2

c. 3

d. 1

31 [Intermediate] • Target: Continuity of Piecewise Functions at Boundaries

Find the value of k that makes $f(x) = 3x + k$ for $x < 6$, and $f(x) = -5x - 8$ for $x \geq 6$, continuous at $x = 6$.

a. -57

b. -20

c. -55

d. -56

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32 [Intermediate] • Target: Implicit Relations and Conic Sections

Find the value of k that makes $f(x) = -9x + k$ for $x < -4$, and $f(x) = 2x - 6$ for $x \geq -4$, continuous at $x = -4$.

a. -49

b. 22

c. -51

d. -50

33 [Intermediate] • Target: Asymptotes and Branches of Hyperbolas

Find the value of k that makes $f(x) = -8x + k$ for $x < 5$, and $f(x) = -2x + 2$ for $x \geq 5$, continuous at $x = 5$.

a. 33

b. -48

c. 31

d. 32

34 [Intermediate] • Target: Evaluating piecewise functions

For the implicit relation $x^2 + y^2 = 25$, find all y-values when $x = 3$.

a. $y = \pm 4$

b. $y = 4$

c. $y = \pm 3$

d. $y = \pm 5$

35 [Intermediate] • Target: Checking continuity of piecewise functions

For the implicit relation $x^2 + y^2 = 1681$, find all y-values when $x = 9$.

a. $y = 40$

b. $y = \pm 40$

c. $y = \pm 41$

d. $y = \pm 9$

36 [Intermediate] • Target: Visualizing a piecewise graph

For $f(x) = 5x - 8$ on $x < 0$ and $f(x) = -3x + 7$ on $x \geq 0$, find the average rate of change of f on $[-1, 2]$.

a. $\frac{14}{3}$

b. $-\frac{14}{3}$

c. $\frac{11}{3}$

d. $\frac{17}{3}$

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37 [Intermediate] • Target: Implicit relations and testing for functions

For $f(x) = -x + 8$ on $x < -3$ and $f(x) = 6x + 7$ on $x \geq -3$, find the average rate of change of f on $[-5, 0]$.

a. $-\frac{6}{5}$

b. $-\frac{11}{5}$

c. $-\frac{1}{5}$

d. $\frac{6}{5}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Implicit differentiation-style reasoning (algebraic slope) • Cross-Lesson: 4.01 Inverse & Composite Functions

A shipping company charges a flat 7 dollars for packages up to 3 kg, plus $\frac{5}{2}$ dollars per kg for anything over 3 kg. Find the cost for a 13 kg package.

a. 32

b. 33

c. 31

d. $\frac{79}{2}$

39 [Challenging] • Target: Modeling with piecewise functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

A shipping company charges a flat 13 dollars for packages up to 2 kg, plus 2 dollars per kg for anything over 2 kg. Find the cost for a 7 kg package.

a. 27

b. 24

c. 23

d. 22

40 [Challenging] • Target: Piecewise functions with three or more pieces • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A shipping company charges a flat 8 dollars for packages up to 2 kg, plus 3 dollars per kg for anything over 2 kg. Find the cost for a 9 kg package.

a. 29

b. 30

c. 28

d. 35

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41 [Challenging] • Target: Evaluating Piecewise Functions • Cross-Lesson: 1.02 Rational Functions & Asymptotes

A tax bracket levies no tax on the first 10,000 dollars, and 15% on income exceeding 10,000 dollars. Which piecewise function $T(x)$ models total tax for income $x \geq 0$?

a. $T(x) = 0$ for $x \leq 10000$, and $T(x) = 0.15(x - 10000)$ for $x > 10000$

b. $T(x) = 0.15x$ for all x

c. $T(x) = 0$ for $x \leq 10000$, and $T(x) = 0.15x$ for $x > 10000$

d. $T(x) = 0$ for $x \leq 10000$, and $T(x) = 1500 + 0.15x$ for $x > 10000$

42 [Challenging] • Target: Continuity of Piecewise Functions at Boundaries • Cross-Lesson: 1.03 Rates of Change & Concavity

Find the value of k that makes $f(x)$ continuous at $x = 3$, where $f(x) = kx + 1$ for $x \leq 3$ and $f(x) = 2x^2 - 5$ for $x > 3$.

a. 4

b. 2

c. 5

d. 13

43 [Challenging] • Target: Continuity of Piecewise Functions at Boundaries • Cross-Lesson: 2.01 Exponential Functions & Growth Models

A function h is defined by $h(x) = |x - 2| + 3$. Where is $h(x)$ continuous and where is it not differentiable?

a. Continuous everywhere; not differentiable at $x = 2$

b. Continuous only for $x \geq 2$; differentiable everywhere

c. Discontinuous at $x = 2$; differentiable elsewhere

d. Continuous and differentiable everywhere on $(-\infty, \infty)$

44 [Challenging] • Target: Implicit Relations and Conic Sections • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Find the center and radius of the circle given by $x^2 + y^2 - 6x + 8y = 0$.

a. Center $(3, -4)$, Radius 5

b. Center $(-3, 4)$, Radius 5

c. Center $(3, -4)$, Radius 25

d. Center $(6, -8)$, Radius 10

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45 [Challenging] • Target: Asymptotes and Branches of Hyperbolas • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Find the equations of the asymptotes for the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{3}{4}x$

b. $y = \pm\frac{4}{3}x$

c. $y = \pm\frac{9}{16}x$

d. $y = \pm\frac{16}{9}x$

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